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The Protocol Selection Problem: PACE-Plan Workload-Class Routing Under DDIL Conditions

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Abstract

P3 established DTN custody transfer as the transport primitive. P4 established WAL + CRDT as the state coherence primitive. P5 established the AI Supervisor as the out-of-path governance layer. What none of these papers specifies is how the substrate selects the correct transport mechanism for each operation in each link condition. The DTN custody layer offers multiple convergence layer adapters, multiple routing algorithms, multiple bundle prioritization schemes, and multiple delivery confirmation modes. The CRDT merge layer offers full-state merge and differential update modes. The contact scheduler offers predictive staging and opportunistic forwarding. Each of these choices has operational consequences: the wrong selection wastes scarce link capacity, violates latency requirements, or degrades state coherence guarantees.

This paper argues that protocol selection at the tactical edge is not a configuration problem but a runtime governance problem. The correct protocol for any given operation depends on three variables that change continuously: the current link conditions (bandwidth, latency, error rate, predicted window duration), the workload class of the data being transmitted (latency-sensitive targeting data has different protocol requirements than bulk historical state synchronization), and the current governance posture (a classification conflict in the pending-custody queue changes the relative priority of the governance policy update bundle). Static protocol configuration — choosing a protocol at deployment and running it until the next maintenance cycle — cannot accommodate this three-variable runtime dependency.

The paper develops a protocol selection architecture in which the AI Supervisor evaluates protocol options against runtime link conditions and workload class requirements, applies governance policy to constrain the selection space, and selects the mechanism that maximizes delivery effectiveness within the governance bounds. This selection is not ad hoc optimization; it is a governed decision recorded in the WAL with the same fidelity as any other Supervisor decision. The §6 Governor Application specifies which protocol selection decisions the Supervisor may make autonomously, which require human governance input, and how the AE² mechanism enforces protocol discipline when link conditions create pressure to use unauthorized shortcuts.

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ABSTRACT

The tactical substrate (EDO-P1) supports multiple communication and coordination protocols, each with distinct cost, latency, capacity, and resilience profiles. The architectural commitment of this paper is to apply the military-doctrine **PACE-plan** framework — Primary / Alternate / Contingency / Emergency — to protocol selection in agentic operation. The agentic system specifies, in advance, which protocol class it uses by default (Primary), which fallback it transitions to under defined degradation conditions (Alternate), which deeper fallback it uses under more severe conditions (Contingency), and which last-resort protocol it uses when other tiers have failed (Emergency). Protocol-level adaptation under load is the operational form of HGC³AE²'s **C**² (Continuity) in the communication dimension.¹²

HOW TO CITE

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1. INTRODUCTION

The cloud-native communication model uses one protocol per concern: HTTPS for API calls, gRPC for inter-service, WebSockets for streaming, S3-style for object storage. Each is operationally adequate in its primary deployment substrate; none is operationally adequate across all of the tactical substrate's degradation profiles.³⁴ The military communications doctrine — **PACE-plan** — has long addressed this through tiered protocol selection: pre-specify what protocol is used by default and what fallbacks are available at increasing levels of degradation.⁵⁶

This paper applies PACE-plan to agentic operation. §2 names five failure modes of single-protocol selection. §3 specifies the PACE-plan structure for agentic protocols. §4 treats protocol selection as continuous. §5 binds to Skipjack. §6 returns to HGC³AE².

1 Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 U.S. Department of the Army, *Field Manual 6-02: Signal Support to Operations* (Washington, DC: Department of the Army, 2019). (Primary military doctrine source; supplemented by refereed analyses below.)

3 Vinton G. Cerf et al., "Delay-Tolerant Networking Architecture," RFC 4838, IETF, 2007.

4 Scott C. Burleigh, Kevin Fall, and Edward J. Birrane III, "Bundle Protocol Version 7," RFC 9171, IETF, 2022.

5 U.S. Department of the Army, *Field Manual 6-02: Signal Support to Operations* (Washington, DC: Department of the Army, 2019). (Primary military doctrine source; supplemented by refereed analyses below.)

6 M. R. Smith and J. P. Hauser, "Tactical Communications Resilience: An Empirical Analysis of PACE-Plan Deployment in Joint Operations," *Military Communications Conference (MILCOM '20)*, 2020, <https://doi.org/10.1109/MILCOM47813.2020.9320990>.



2. FAILURE MODES OF SINGLE-PROTOCOL SELECTION

Brittleness at degradation thresholds. The chosen protocol works until a threshold is crossed, then fails entirely; no graceful degradation.

Latency cliff. Protocols that retry under congestion can produce latency cliffs that crash dependent timing-sensitive workloads.

Bandwidth amplification. Some protocols carry substantial framing overhead that becomes a binding constraint when bandwidth is scarce.

Compromise-mode persistence. When the chosen protocol is compromised (security failure, vendor compromise, capability removal), there is no architecturally specified fallback.

Authority confusion under fallback. Ad-hoc fallback decisions made under stress produce authority ambiguity about who authorized the protocol transition.

3. PACE-PLAN FOR AGENTIC PROTOCOLS

3.1 PRIMARY

The default protocol class — typically full-bandwidth, low-latency, full-feature. For agentic operation under cloud-native conditions, the Primary tier is HTTPS/gRPC/WebSockets/S3-style with the cloud-native stack.

3.2 ALTERNATE

The first fallback — typically reduced-bandwidth or higher-latency but functional. Triggered when Primary's operating envelope is exceeded. For agentic operation, the Alternate tier may include DTN bundles (EDO-P3) over reduced-capacity links, with explicit understanding that the operating cadence will slow.⁷

3.3 CONTINGENCY

The deeper fallback — typically minimal-bandwidth, asynchronous-only, governance-essential-only. Triggered when Alternate is unavailable. For agentic operation, the Contingency tier carries only governance and persistence-essential cargo; work-in-flight may pause.

3.4 EMERGENCY

The last-resort protocol — typically out-of-band, manually keyed, governance-only. Triggered when all other tiers have failed. For agentic operation, the Emergency tier preserves operator-substrate communication for envelope re-authorization but not work execution.

3.5 TRANSITIONS

Each transition between PACE levels is governance-defined: the conditions that trigger the transition, the protocol class adopted at the new level, the implications for in-flight work, and the

⁷ EDO-P3 *When the Link Is the Variable* promotion (in concurrent drafting; yks-build#95).



conditions for returning to a higher tier. Transitions are not auto-tuned; they are specified by the governance layer at envelope-authorization time.⁸⁹

4. PROTOCOL SELECTION AS CONTINUOUS PROCESS

Three phases. **Monitoring** runs continuously: link state, capacity availability, latency variance, error rate. **Evaluation** runs at every meaningful change: does the current PACE level remain appropriate? **Transition** runs when evaluation indicates: explicit transition between PACE levels with operator notification.

The continuous-process framing is the operational form of HGC³AE²'s C² for communication: continuity through tiered fallback rather than through single-protocol resilience.

5. SKIPJACK COUPLING

Skipjack mechanisms operate over each PACE tier with tier-appropriate adaptations.¹⁰ **Context integrity** is enforced at every tier; lower tiers carry less context per bundle but the integrity discipline is uniform. **Structured routing** operates within the protocol class's capacity. **HITL control points** fire at PACE-level transitions (the transition itself is a governance-significant event). **Iterative feedback** closes asynchronously across tier transitions.

6. IMPLICATIONS AND CONCLUSION

The implications extend to architecture (each PACE tier requires explicit protocol-class specification), operational doctrine (operators must train on transitions), and tooling (PACE-level transition logic must be testable and verifiable).

The core claim: **PACE-plan applied to agentic protocols preserves operational continuity under tiered degradation; without explicit tiered protocols, the operating model is brittle at degradation thresholds.**

Returned to HGC³AE² letter by letter:

H — Human-governed. PACE-plan structure is governance-authored.

G — Curated Context. Each tier carries curated context appropriate to its capacity.

C¹ — Cybersecurity. Each tier has its own security profile; transitions are themselves security-significant events.

C² — Continuity. Continuity through tiered fallback is the paper's load-bearing commitment.

C³ — Coherence. Tier transitions preserve coherence through governance-defined state propagation.

A — Agentially Engineered. Tier evaluation is agentic within governance-defined thresholds.

8 Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

9 Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

10 Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.



E — Executed. Continuous monitoring and tier evaluation.

The exponent — the closure — is the monitoring-evaluation-transition loop.

The remaining EDO papers (workload class, classification, accreditation, interoperability, deployment) operate over the protocol-tier substrate this paper specifies.¹¹¹²

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Drafting state: v1.0-preprint draft. PR count: 12 inline (FM 6-02 primary + Smith MILCOM · Cerf RFC · Burleigh RFC · Singh IEEE Comm · Kreutz PIEEE · Guerin Springer · Bonaventure IC · Hollnagel Ashgate · Mogul SIGCOMM · Beyer SRE · Hinrichs SoCC).

11 EDO-P7 *Workload Class Problem* promotion (in concurrent drafting; yks-build#106).

12 EDO-P11 CAPSTONE promotion (in concurrent drafting; yks-build#110).